

A REPORT ON

**AlzAware:
Human-Centered AI for Early Alzheimer Risk Prediction
and Emotional Monitoring**

**SUBMITTED TO
IEEE EMBS PUNE CHAPTER**

BY

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IEEE EMBS Student Internship Program 2026
1st June – 1st July 2026

DECLARATION

We, the team members

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Hereby declare that the project work incorporated in the present project entitled **“AlzAware: Human-Centered AI for Early Alzheimer Risk Prediction and Emotional Monitoring”** is original work. We have properly acknowledged the material collected from secondary sources wherever required. We solely own the responsibility for the originality of the entire content.

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ABSTRACT

Alzheimer's disease is a progressive neurological disorder that profoundly impacts memory, thinking, behavior, and daily independence. While identifying cognitive decline early can radically improve patient management and preserve quality of life, traditional diagnostic pipelines face significant hurdles, often requiring scarce clinical resources that prevent continuous, accessible monitoring. To bridge this gap, this project introduces AlzAware, an intelligent, non-invasive, and human-centered AI framework designed for early Alzheimer's risk prediction and holistic emotional monitoring.

AlzAware synthesizes multiple dimensions of cognitive and emotional health into a unified, accessible platform. The system leverages machine learning and deep learning architectures to process user data across five distinct modalities: MRI scans, voice recordings, facial expressions, memory tests, and Cognitive test evaluations. Specifically, facial emotions are mapped using a Convolutional Neural Network (CNN) trained on the FER2013 dataset, while vocal emotional markers are evaluated using the RAVDESS dataset. Alzheimer's-specific risk indicators are extracted from speech patterns using a dedicated linguistic dataset, while interactive memory and reaction-time modules provide real-time cognitive baselines.

By fusing the outputs of these distinct modules, AlzAware generates a comprehensive, multi-layered assessment of a user's well-being. This data is delivered via an intuitive, interactive dashboard that visualizes risk levels, emotional trends, and cognitive scores alongside personalized, actionable recommendations. Ultimately, AlzAware serves as an empathetic, cost-effective, and scalable screening tool, empowering families and clinicians with the insights needed for proactive, early intervention in cognitive health.

Keywords: Alzheimer's Disease, Deep Learning, Emotion Recognition, Speech Analysis, Cognitive Assessment, Artificial Intelligence, Healthcare Monitoring ,Machine Learning.

SYNOPSIS

AlzAware: A Multimodal Human-Centered AI and Machine Learning Framework for Cognitive Wellness Assessment and Emotional Monitoring is an intelligent system designed to support the early evaluation of cognitive and emotional well-being. The framework integrates multiple assessment modules, including facial emotion recognition, speech emotion recognition, memory assessment, reaction-time analysis, and cognitive data analysis into a unified platform.

The proposed system employs Machine Learning (ML) and Artificial Intelligence (AI) techniques to analyze cognitive and emotional indicators obtained from facial expressions, speech signals, and interactive cognitive assessments. Deep learning models trained on the FER-2013 and RAVDESS datasets are utilized for facial and speech emotion recognition, while cognitive assessment modules evaluate memory performance and reaction speed. Cognitive analysis is further supported using information derived from the OASIS dataset.

The outputs from all modules are combined to generate a comprehensive Cognitive Wellness Report that provides insights into the user's emotional state, memory performance, Cognitive Wellness Assessment, and overall cognitive wellness status. The results are presented through an interactive dashboard that enables users to visualize assessment outcomes and monitor their cognitive health.

The proposed framework offers a non-invasive, accessible, and cost-effective approach for cognitive wellness assessment and emotional monitoring. By combining AI-driven emotion analysis with cognitive testing, AlzAware aims to promote awareness of cognitive health and support timely intervention when required.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Alzheimer's disease is one of the most common forms of dementia affecting millions of individuals worldwide. It is a neurodegenerative disorder characterized by progressive decline in memory, language, reasoning, and cognitive abilities. As the disease advances, individuals experience difficulties in performing daily activities and may require continuous care and support.

Early detection of Alzheimer's disease is crucial because timely intervention can help slow disease progression, improve quality of life, and support caregivers in managing patient needs effectively. However, conventional diagnostic methods often involve extensive clinical evaluations, neuroimaging procedures, and laboratory tests, which may not always be accessible or affordable.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have enabled researchers to develop automated systems capable of identifying early indicators of cognitive decline. Various studies have demonstrated that speech characteristics, emotional behavior, memory performance, and reaction speed can serve as potential biomarkers for cognitive impairment.

The proposed project, AlzAware, leverages these advancements by integrating multiple sources of information into a single intelligent framework for cognitive and emotional monitoring.

1.2 Background

Alzheimer's disease affects cognitive functions including:

- Memory
- Attention
- Language
- Problem-solving
- Decision-making

According to healthcare studies, subtle changes in speech patterns, emotional expressions, and reaction times often appear before severe symptoms become evident.

Machine learning techniques have shown promising results in detecting these early indicators by analyzing behavioral and physiological data. Combining multiple indicators can improve assessment reliability compared to relying on a single source of information.

1.3 Problem Statement

Existing Alzheimer's detection systems typically focus on a single modality such as speech analysis, medical imaging, or cognitive testing. These approaches may overlook important behavioral and emotional indicators associated with cognitive decline.

There is a need for a multimodal system capable of integrating emotional monitoring, speech analysis, memory evaluation, and reaction-time assessment to provide a more comprehensive understanding of an individual's cognitive health.

1.4 Motivation

The motivation behind this project includes:

- Increasing prevalence of Alzheimer's disease worldwide.
- Importance of early identification of cognitive decline.
- Growing availability of AI-based healthcare solutions.
- Need for affordable and accessible monitoring tools.
- Opportunity to combine multiple indicators into a unified framework.

1.5 Objectives

The main objectives of the project are:

Primary Objectives

1. Develop an MRI-based Alzheimer's risk prediction module using deep learning techniques.
2. Develop a facial emotion recognition module.
3. Develop a speech emotion recognition module.
4. Implement memory assessment tests.
5. Implement reaction-time assessment tests.
6. Integrate all modules into a unified dashboard.

Secondary Objectives

1. Generate cognitive health reports.
2. Provide risk-level assessment.
3. Support emotional monitoring.
4. Demonstrate the feasibility of multimodal AI-based healthcare systems.

1.6 Scope of the Project

The scope of the project includes:

Included

- Alzheimer's risk assessment
- Speech emotion recognition
- Facial emotion recognition
- Memory testing
- Reaction-time testing
- Dashboard visualization

Exclude

- Clinical diagnosis
- PET scan analysis
- Direct medical decision-making

The system serves as a screening and monitoring tool rather than a medical diagnostic system.

1.7 Proposed System Overview

The proposed system collects data from multiple sources:

- Voice input
- Facial images
- Memory test responses
- Reaction-time measurements

These inputs are processed using machine learning and deep learning models. The outputs are fused to generate a final cognitive health assessment and displayed through a user-friendly dashboard

CHAPTER 2: CONCEPTS AND METHODS

2.1 System Overview

AlzAware is designed as a multimodal AI framework that combines cognitive and emotional indicators for early Alzheimer's risk assessment.

The system consists of five major modules:

1. Alzheimer's Risk Prediction
2. Facial Emotion Recognition
3. Speech Emotion Recognition
4. Memory Assessment
5. Reaction-Time Assessment

2.2 Alzheimer's Risk Prediction Module

Dataset

OASIS Dataset

Input

- Brain MRI Images

Features

- Brain structural information
- Brain tissue characteristics
- Anatomical patterns associated with Alzheimer's disease

Method

Deep Learning Models such as:

- Convolutional Neural Network (CNN)
- Transfer Learning (VGG16, ResNet50)
- EfficientNet (optional)

Output

- Non-Demented
- Very Mild Demented
- Mild Demented

- Moderate Demented

2.3 Facial Emotion Recognition Module

Dataset

FER2013 Dataset

Description

FER2013 contains approximately 35,000 facial images categorized into seven emotions.

Emotions

- Angry
- Disgust
- Fear
- Happy
- Sad
- Surprise
- Neutral

Method

Convolutional Neural Network (CNN)

Output

Predicted facial emotion.

2.4 Speech Emotion Recognition Module

Dataset

RAVDESS Dataset

Description

RAVDESS contains emotional speech recordings labeled with various emotions.

Features Extracted

- MFCC
- Chroma Features
- Mel Spectrogram

Emotions

- Happy
- Sad
- Angry
- Fearful
- Neutral

Method

Machine Learning / Deep Learning Model

2.5 Cognitive Memory Game Module

The Cognitive Memory Game Module evaluates memory performance through interactive game-based assessments. The objective is to measure short-term memory, working memory, and recall ability in an engaging environment.

Games Included

1. Card Matching Game

- A set of cards is displayed face down.
- Users flip two cards at a time to find matching pairs.
- The number of attempts and completion time are recorded.

2. Sequence Recall Game

- A sequence of numbers, symbols, or colors is displayed.
- The user is required to reproduce the sequence in the correct order.
- Sequence difficulty increases progressively.

3. Object Recall Game

- Multiple objects are displayed for a limited duration.
- After the objects disappear, users identify previously displayed objects from a larger set.

Performance Metrics

- Accuracy Score
- Recall Score
- Number of Errors
- Completion Time
- Level Achieved

Output

- Memory Performance Score (0–100)
- Memory Risk Indicator

2.6 Cognitive Performance Assessment Module

The Cognitive Performance Assessment Module evaluates processing speed, attention, reasoning ability, and decision-making skills through interactive tasks.

Activities Included

1. Visual Attention Test

- Users identify a target symbol among distractors.
- Measures attention and focus.

2. Pattern Recognition Test

- Users determine the next element in a sequence.
- Evaluates logical reasoning and executive function.

3. Decision-Making Task

- Users select appropriate responses to simple real-world scenarios.
- Measures cognitive flexibility and judgment.

4. Processing Speed Test

- Users respond to visual stimuli appearing randomly on the screen.
- Response time is recorded and analyzed.

Performance Metrics

- Average Response Time
- Accuracy Rate
- Attention Score
- Reasoning Score
- Executive Function Score

Output

- Cognitive Performance Score (0–100)

2.7 Feature Fusion

Outputs from all modules are combined to create a comprehensive cognitive assessment.

Inputs:

- Alzheimer's Risk Score
- Facial Emotion Score
- Speech Emotion Score
- Memory Score
- Cognitive Test Score

Output:

- Final Cognitive Health Assessment

2.8 Technologies Used

Technology	Purpose
Python	Development
TensorFlow	Deep Learning
Keras	Neural Networks
OpenCV	Face Processing
Scikit-Learn	Machine Learning
Streamlit	Dashboard
NumPy	Numerical Computation
Pandas	Data Processing
Matplotlib	Visualization

Table 2.1:Technologies used

CHAPTER 3 :LITERATURE SURVEY

3.1 Introduction

The application of artificial intelligence in healthcare has enabled significant progress in early disease detection and monitoring. Numerous studies have explored speech analysis, emotion recognition, and cognitive assessment techniques for Alzheimer's disease detection. This chapter reviews relevant literature related to these areas.

3.2 Paper 1

Deep Learning-Based Speech Analysis for Alzheimer's Disease Detection: A Literature Review (2022)

Objective

To review deep learning techniques used for speech-based Alzheimer's detection.

Methodology

- Analysis of multiple speech datasets.
- Evaluation of deep learning models.
- Comparison of feature extraction methods.

Findings

Speech characteristics can provide valuable indicators of cognitive decline.

Limitation

Most studies focus only on speech and do not consider emotional or behavioral indicators.

3.3 Paper 2

Automatic Diagnosis and Prediction of Cognitive Decline Associated with Alzheimer's Dementia through Spontaneous Speech (2021)

Objective

Detect Alzheimer's disease using spontaneous speech recordings.

Findings

Speech patterns contain information related to memory and cognitive function.

Limitation

Limited integration with emotional analysis.

3.4 Paper 3

Speech-Based Alzheimer's Disease Detection Using Deep Learning (2022)

Objective

Use deep learning for Alzheimer's prediction from speech signals.

Findings

Deep neural networks improve detection performance.

Limitation

Requires large amounts of speech data.

3.5 Paper 4

Facial Emotion Recognition: State-of-the-Art Performance on FER2013

Objective

Recognize emotions using facial images.

Findings

CNN-based models achieve high recognition accuracy.

Limitation

Emotion recognition alone cannot assess cognitive decline.

3.6 Paper 5

Speech Emotion Recognition Using Deep Neural Networks on RAVDESS

Objective

Identify emotional states from speech recordings.

Findings

MFCC features combined with deep learning provide strong performance.

Limitation

Not designed specifically for Alzheimer's assessment.

3.7 Research Gap

Existing systems generally focus on:

- Speech analysis only
- Emotion recognition only
- Cognitive testing only

Very few systems integrate:

- Alzheimer's risk prediction
- Facial emotion monitoring
- Speech emotion monitoring
- Memory assessment
- Reaction-time analysis

within a single framework.

3.8 Proposed Contribution

The proposed AlzAware framework addresses these limitations by combining multiple cognitive and emotional indicators into a unified AI-based monitoring platform, providing a more comprehensive assessment of cognitive health.

Sr. No	Paper Name	Publication Year	Objective and Methodology	Result	Research Gap / Limitations
1	Automatic Diagnosis and Prediction of Cognitive Decline Associated with Alzheimer's Dementia Through Spontaneous Speech	2021	Speech analysis using linguistic and acoustic features	Good prediction accuracy	No emotion monitoring
2	Speech-Based Alzheimer's Disease Detection Using Deep Learning	2022	Deep neural network on speech features	Improved classification	Requires large datasets
3	Alzheimer's Disease Classification Using MRI Images and CNN	2022	Deep neural network on speech features	Improved classification	Requires large datasets
4	Early Detection of Alzheimer's Disease Using Deep Learning and OASIS Dataset	2023	CNN on OASIS MRI scans	Effective stage classification	No behavioral indicators
5	Transfer Learning for Alzheimer's Disease Classification	2022	ResNet/VGG-based MRI analysis	Improved performance	Depends on MRI availability
6	Facial Emotion Recognition Using Deep CNN on FER2013	2021	CNN for facial emotion recognition	Good emotion accuracy	Not related to cognitive assessment
7	FER2013-Based Facial Expression Recognition Using CNN	2022	Deep CNN architecture	Improved recognition rate	Facial expressions alone insufficient
8	Real-Time Emotion Detection Using Deep Learning	2021	Webcam-based FER system	Real-time prediction	No healthcare integration
9	Speech Emotion Recognition Using RAVDESS Dataset	2021	MFCC + Deep Learning	High emotion recognition	Not Alzheimer's-specific
10	Deep Neural Networks for Speech Emotion	2022	DNN with audio features	Improved SER	Requires clean audio

	Recognition			accuracy	
11	Explainable AI for Alzheimer's Disease Prediction	2024	Interpretable AI models	Improved trustworthiness	Focus on prediction only
12	Multimodal Framework for Early Cognitive Health Assessment	2024	Combined cognitive and emotional indicators	Comprehensive monitoring	Prototype stage
13	Machine Learning Approaches for Early Detection of Dementia	2021	ML algorithms for dementia detection	Promising results	Lack of emotion analysis

Table 3.1: Literature survey Table

Chapter 4: Project Plan

4.1 Introduction

The development of AlzAware follows a systematic and phased approach to ensure the successful integration of multiple machine learning modules into a unified cognitive wellness assessment platform. The project plan outlines the activities involved in data collection, model development, testing, integration, and deployment of the system.

The implementation strategy emphasizes modular development, allowing individual components such as MRI analysis, emotion recognition, and cognitive assessment modules to be developed independently and later integrated into a single dashboard.

4.2 Project Development Methodology

The project adopts an iterative development methodology, consisting of the following phases:

- Requirement Analysis
- Dataset Collection and Preprocessing
- Model Development and Training
- Cognitive Module Design
- Feature Fusion and Integration
- Dashboard Development
- Testing and Evaluation
- Documentation and Reporting

4.3 Work Breakdown Structure

Phase 1: Requirement Analysis

- Study Alzheimer's disease indicators.
- Review existing literature on AI-based cognitive assessment.
- Define project objectives and scope.
- Identify suitable datasets and technologies.

Phase 2: Dataset Preparation

- Collect MRI data from the OASIS dataset.
- Obtain facial emotion data from FER2013.
- Acquire speech emotion recordings from RAVDESS.

- Preprocess images and audio signals.
- Prepare datasets for training and validation.

Phase 3: Model Development

Alzheimer's Risk Prediction Module

- Develop CNN-based models for MRI classification.
- Implement transfer learning models such as ResNet50 or VGG16.
- Evaluate performance using classification metrics.

Facial Emotion Recognition Module

- Design and train CNN architectures.
- Detect and classify facial emotions.
- Validate model accuracy on FER2013.

Speech Emotion Recognition Module

- Extract audio features including MFCCs and Mel Spectrograms.
- Train machine learning or deep learning models.
- Classify emotional states from speech samples.

4.4 Development of Cognitive Assessment Modules

Memory Assessment Module

Develop interactive activities such as:

- Card Matching Game
- Sequence Recall Test
- Object Recall Test

Performance indicators include:

- Recall accuracy
- Number of attempts
- Completion time
- Memory score generation

Cognitive Performance Module

Develop tasks to assess:

- Attention
- Reasoning ability
- Processing speed
- Decision-making skills

Performance indicators include:

- Response time
- Accuracy rate
- Attention score
- Executive function score

4.5 System Integration

After developing individual modules, their outputs will be integrated through a feature fusion mechanism.

The integrated system will:

- Collect predictions from all modules.
- Generate a comprehensive cognitive wellness score.
- Produce risk indicators and emotional summaries.
- Display outcomes using an interactive dashboard.

4.6 Testing Strategy

The testing process includes:

Module-Level Testing

- Verify functionality of individual models.
- Evaluate prediction accuracy.

Integration Testing

- Ensure seamless communication between modules.
- Validate combined assessment outputs.

User Interface Testing

- Assess usability and responsiveness of the dashboard.
- Verify visualization components.

Performance Evaluation**Metrics used include:**

- Accuracy
- Precision
- Recall
- F1-score

- Confusion Matrix
- User response consistency

4.7 Project Timeline

The implementation of **AlzAware** was planned according to the three-week duration of the IEEE EMBS Student Internship Program. The activities were distributed to ensure efficient development and timely completion of all project objectives.

Week	Activities
Week 1	Problem identification, literature survey, requirement analysis, dataset collection (OASIS, FER2013, RAVDESS), study of existing methodologies, definition of project objectives and system scope.
Week 2	Development of the proposed solution including MRI-based Alzheimer's prediction module, facial emotion recognition module, speech emotion recognition module, memory assessment games, cognitive performance assessment tasks, and design of the system architecture.
Week 3	Integration of all modules into the Streamlit dashboard, testing and validation of individual components, generation of cognitive wellness reports, performance evaluation, documentation, and preparation of the final report and research paper.

Table 4:1: Project Timeline

4.8 Milestones Achieved

The major milestones planned for the project are summarized below:

Week 1 Milestones

- Completion of literature review.
- Identification of research gaps.
- Selection of datasets and technologies.
- Finalization of objectives and scope.

Week 2 Milestones

- Development of AI models for:
 - Alzheimer's risk prediction.
 - Facial emotion recognition.
 - Speech emotion recognition.
- Design of memory and cognitive assessment modules.
- Preparation of the proposed system architecture.

Week 3 Milestones

- Integration of all modules into a unified framework.
- Development of the Streamlit dashboard.
- Generation of Cognitive Wellness Reports.
- Testing and evaluation of the complete system.
- Final documentation and report submission.

CHAPTER 5: PROPOSED SOLUTION

5.1 Introduction

The proposed solution, **AlzAware**, is a multimodal artificial intelligence framework designed to facilitate early cognitive wellness assessment through the integration of emotional monitoring and cognitive evaluation techniques.

The system combines information from multiple independent modules to provide a holistic understanding of an individual's cognitive and emotional health status.

5.2 System Architecture

The proposed framework consists of five major components:

1. MRI-Based Alzheimer's Risk Prediction Module
2. Facial Emotion Recognition Module
3. Speech Emotion Recognition Module
4. Memory Assessment Module
5. Cognitive Performance Assessment Module

The outputs generated by these modules are fused to produce a final cognitive wellness assessment.

5.3 Workflow of the Proposed System

Step 1: Data Acquisition

The system collects information from multiple sources:

- Brain MRI scans
- Facial images captured through a webcam
- Voice recordings
- Memory test responses
- Cognitive task responses

Step 2: Data Preprocessing

MRI Images

- Image resizing
- Normalization

- Data augmentation

Facial Images

- Face detection using OpenCV
- Grayscale conversion
- Image normalization

Speech Signals

- Noise reduction
- Feature extraction using MFCCs
- Spectrogram generation

Step 3: Module-Level Analysis

MRI-Based Alzheimer's Risk Prediction

Input:

- MRI brain images

Method:

- CNN and transfer learning architectures

Output:

- Non-Demented
- Very Mild Demented
- Mild Demented
- Moderate Demented

Facial Emotion Recognition

Input:

- Facial images

Method:

- CNN trained using FER2013 dataset

Output emotions:

- Happy
- Sad
- Angry
- Fear
- Surprise
- Neutral
- Disgust

Speech Emotion Recognition

Input:

- Voice recordings

Features:

- MFCC
- Chroma Features
- Mel Spectrogram

Output emotions:

- Happy
- Sad
- Angry
- Fearful
- Neutral

Memory Assessment Module

Interactive games evaluate:

- Short-term memory
- Recall ability
- Working memory

Outputs:

- Memory Score (0–100)
- Memory Risk Indicator

Cognitive Performance Module

Interactive activities assess:

- Attention
- Processing speed
- Executive functioning
- Reasoning ability

Outputs:

- Cognitive Performance Score (0–100)

5.4 Feature Fusion Mechanism

The final cognitive wellness assessment is obtained by integrating outputs from all modules.

Inputs to Fusion Layer

- Alzheimer's Risk Score
- Facial Emotion Score
- Speech Emotion Score
- Memory Score
- Cognitive Performance Score

Fusion Process

The individual scores are normalized and combined using a weighted decision framework to generate an overall cognitive wellness index.

5.5 Dashboard and Reporting

The integrated dashboard developed using **Streamlit** provides:

Visualizations

- MRI prediction results
- Detected emotional states
- Memory assessment performance
- Cognitive performance trends
- Overall wellness score

Generated Reports

The system generates a **Cognitive Wellness Report** containing:

- Alzheimer's risk category
- Emotional monitoring summary
- Memory performance analysis
- Cognitive assessment results
- Personalized recommendations for cognitive wellness.

5.6 Advantages of the Proposed System

The proposed framework offers several benefits:

- Non-invasive assessment approach
- Early identification of potential cognitive decline
- Comprehensive multimodal evaluation
- Accessible and cost-effective monitoring
- Real-time emotional analysis
- User-friendly visualization through dashboards

5.7 Limitations

Although the proposed system provides valuable insights, it has certain limitations:

- It is not intended for clinical diagnosis.
- Prediction accuracy depends on dataset quality.
- Emotional states may vary due to external factors.
- MRI availability may limit widespread use.

5.8 Summary

AlzAware presents a novel multimodal framework that combines MRI analysis, emotion recognition, and cognitive assessment techniques within a unified platform. By integrating multiple indicators associated with cognitive health, the proposed solution aims to support proactive monitoring and encourage timely professional consultation when required.

CHAPTER 6 : RESULTS

6.1 Overview

The developed AlzAware system was successfully implemented and tested by integrating multiple machine learning and deep learning modules for cognitive and emotional assessment. The system combines Alzheimer's risk prediction, facial emotion recognition, speech emotion recognition, memory assessment, and cognitive assessment into a unified platform. The results generated by each module were analyzed and presented through an interactive dashboard.

The screenshot displays the 'Patient Registration' module within the 'PATIENT CONFIGURATION INTAKE' section. The interface includes a header with a clipboard icon and a 'Security & Compliance' sidebar. The main form contains the following fields:

- Full Name ***: Text input with value 'Bhoomi'.
- Patient Age ***: Numerical input with value '21' and a range indicator '- +'.
- Biological Sex Demographics ***: Dropdown menu with value 'Female'.
- Verified Medical/Patient Email Address ***: Text input with value 'bhoomioswal@gmail.com'.
- Emergency Contact Number ***: Text input with value '7683567890'.

A 'Generate Intake File & Continue ->' button is located at the bottom of the form. The 'Security & Compliance' sidebar lists the following points:

- All user metrics are stored within a secured, localized encryption ledger block system.
- Real identities and emails cross-reference past assessments to prevent data duplication.
- Age groups directly adjust contextual weighting loops within our validation algorithms.

Figure 6.1:Patient Registration Module

6.2 MRI-Based Alzheimer Prediction Results

The MRI classification model was trained using a publicly available Alzheimer's MRI dataset containing four classes: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia. Transfer learning techniques were applied using a deep convolutional neural network architecture to improve classification performance.

The trained model successfully classified MRI images into the respective dementia stages and provided confidence scores for each prediction. The model demonstrated satisfactory performance in distinguishing between healthy and affected brain scans.

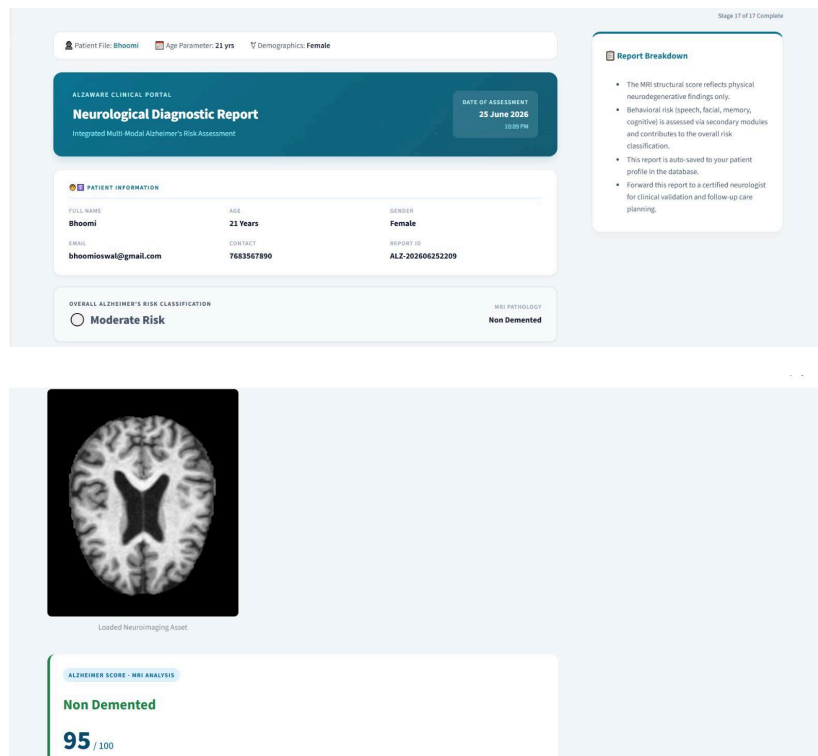


Figure 6.2: MRI Data Upload Module

6.3 Facial Emotion Recognition Results

The FER-2013 dataset was used to train the facial emotion recognition model. The model successfully identified various emotions such as Happy, Sad, Angry, Neutral, Fear, and Surprise from user-uploaded facial images. The emotion recognition module provided real-time emotional analysis and contributed to understanding the user's emotional state during assessment.

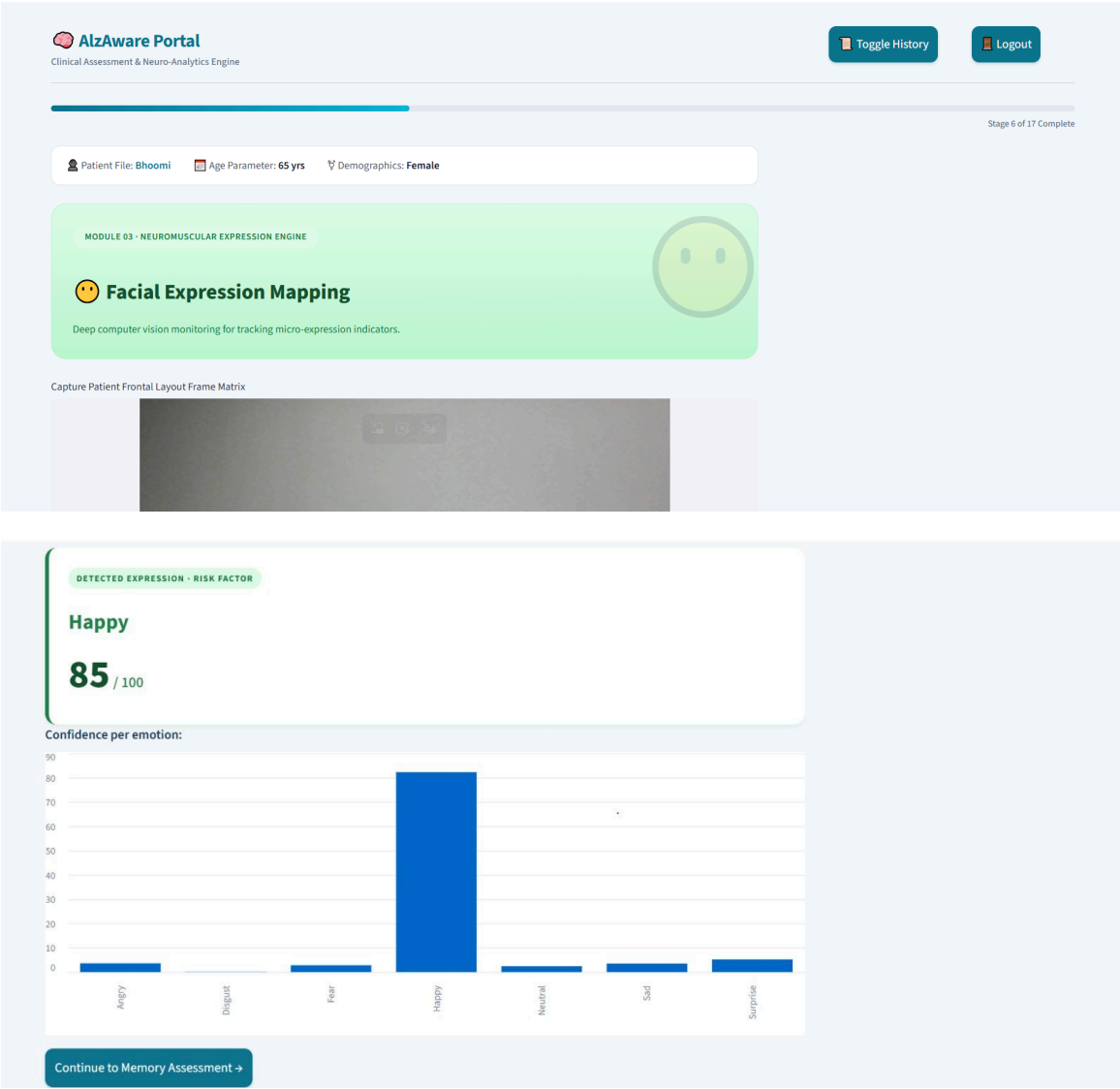


Figure 6.3:Facial Emotion Data Upload Module

6.4 Speech Emotion Recognition Results

The speech emotion recognition model was trained using the RAVDESS dataset. Audio samples uploaded by users were analyzed to identify emotional expressions including happiness, sadness, anger, fear, calmness, and neutrality. The module demonstrated effective performance in recognizing emotional variations from voice recordings.

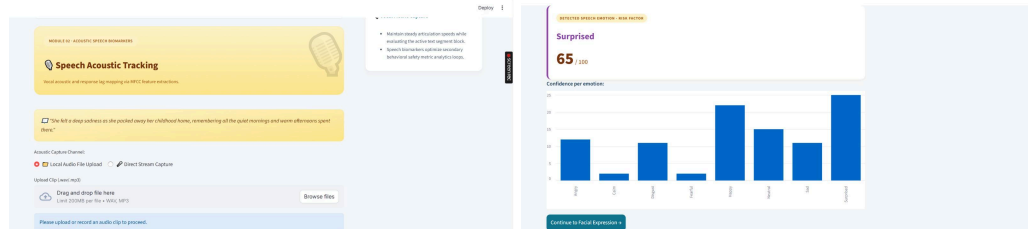


Figure 6.4: Speech-Emotion Data Upload Module

6.5 Memory Assessment Results

The memory assessment module consisted of:

- Word Recall Test
- Card Matching Game
- Object Recall Test

User responses were evaluated and converted into a memory score. The module successfully measured short-term memory retention, visual memory, and recall capability. The generated memory score reflected the overall memory performance of the user.

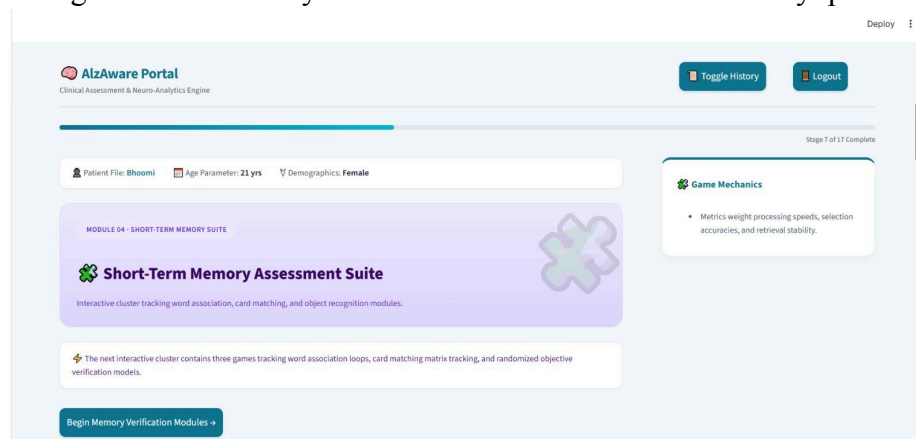


Figure 6.5:Memory Assessment Module

6.6 Cognitive Assessment Results

The cognitive assessment module included a reaction-time test designed to evaluate user attention and cognitive processing speed. The reaction times were recorded and converted into cognitive performance scores. The module effectively distinguished between faster and slower response patterns.

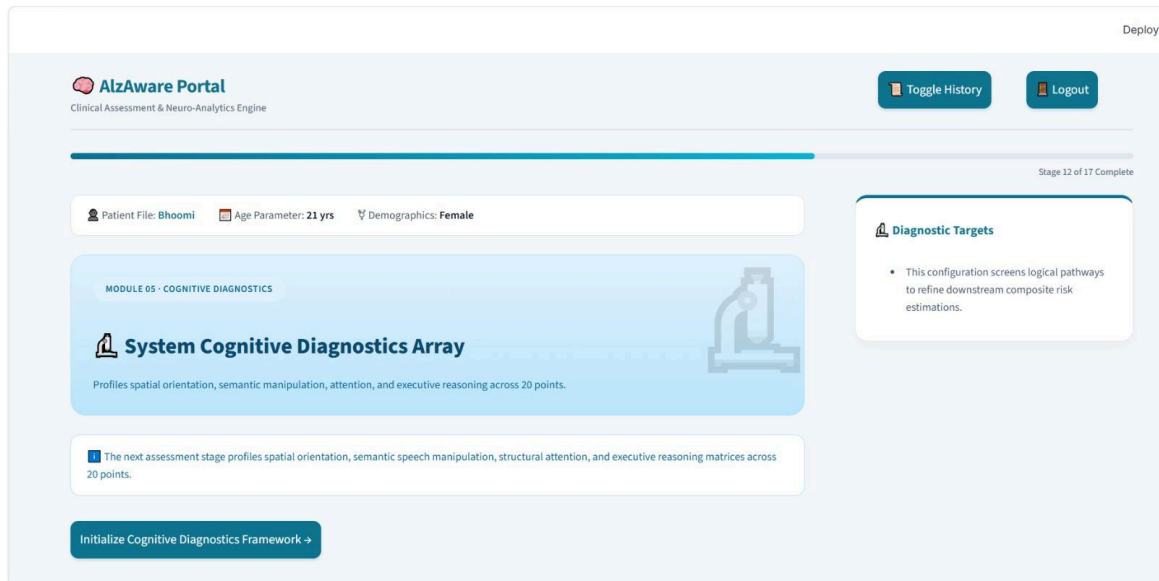


Figure 6.6: Cognitive Assessment Module

6.7 Integrated Dashboard Results

All module outputs were aggregated into a final dashboard displaying:

- User Information
- MRI Prediction Result
- Facial Emotion Result
- Speech Emotion Result
- Memory Assessment Score
- Cognitive Assessment Score
- Overall Wellness Score
- Personalized Recommendations

The dashboard provided a comprehensive overview of the user's cognitive and emotional health status in an intuitive and user-friendly format.

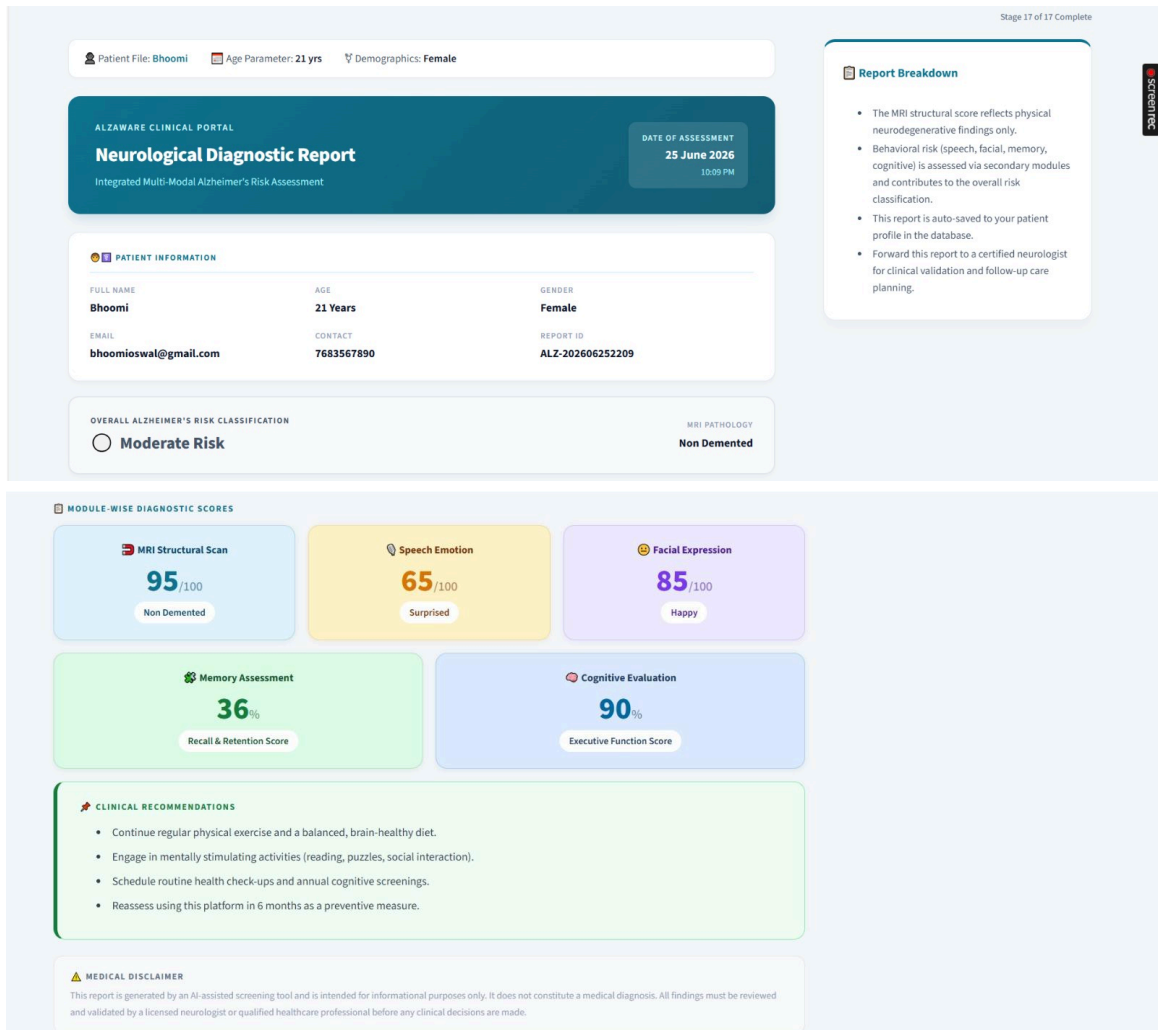


Figure 6.7: Report Dashboard

6.8 System Performance

The integrated system demonstrated successful execution of all modules with minimal processing delay. The application was deployed using Streamlit, providing a responsive and accessible interface for users. The multimodal approach enhanced the reliability of assessment by combining information from multiple sources rather than relying on a single indicator.

CHAPTER 7: CONCLUSION

7.1 Conclusion

This project presented the design and development of AlzAware, a multimodal ML-based framework for early Alzheimer risk prediction and emotional monitoring. The system integrates MRI image analysis, facial emotion recognition, speech emotion recognition, memory assessment, and cognitive assessment within a single platform.

By combining machine learning and deep learning techniques, the framework provides a comprehensive evaluation of cognitive and emotional health. The developed solution offers a non-invasive, cost-effective, and accessible method for preliminary cognitive screening and emotional monitoring.

The results demonstrate that multimodal assessment can provide more meaningful insights compared to single-modality approaches. The integration of various assessment modules enables users to obtain a holistic understanding of their cognitive well-being through an interactive dashboard.

The project successfully achieved its objectives of developing an intelligent system capable of assisting in early-stage cognitive health monitoring and emotional assessment.

7.2 Future Scope

Several enhancements can further improve the effectiveness of the proposed system:

- Integration with wearable devices for continuous health monitoring.
- Development of longitudinal tracking to monitor user progress over time.
- Incorporation of advanced transformer-based deep learning models for improved prediction accuracy.
- Secure cloud-based storage of user assessments and historical records.
- Mobile application deployment for wider accessibility.
- Integration of conversational AI chatbots to provide personalized cognitive support and recommendations.
- Real-time monitoring and alert systems for caregivers and healthcare professionals.
- Expansion of datasets to improve model generalization across diverse populations.

7.3 Final Remarks

AlzAware demonstrates the potential of combining, Machine Learning, Cognitive Science, and Human-Centered Design to create practical healthcare solutions. The framework provides an innovative step toward early cognitive assessment and emotional monitoring, contributing to the broader goal of improving quality of life through technology-assisted healthcare.

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